

A.7 | Ultra-rel:  $kT \gg mc^2$   
 $\frac{1}{2} m v_{rms}^2 \approx \frac{N}{3} kT$

$E \rightarrow E \approx pc \leftarrow (E^2 = (pc)^2 + (mc^2)^2)$

Non-Rel:  $kT \ll mc^2$   $E = mc^2 + \frac{p^2}{2m}$

$\rightarrow \left( e^{\frac{mc^2}{kT}} \cdot e^{\frac{p^2}{2mkT}} \right)^{-1}$   
 $\frac{mc^2}{kT} \gg 1$   $\uparrow$   $\pm 1$   $\gg 1$

$\frac{1}{e^{\frac{mc^2}{kT}} e^{\frac{p^2}{2mkT}} \pm 1} \rightarrow e^{-\frac{mc^2}{kT}} e^{-\frac{p^2}{2mkT}}$   
 $\pm 1$ : can ignore

Ultra-rel:  $kT \gg mc^2$   $\Rightarrow E \approx pc$

$n = \frac{g}{(2\pi\hbar)^3} \int_0^\infty \frac{4\pi p^2}{e^{\frac{pc}{kT}} \pm 1} dp$

$y = \frac{pc}{kT} \Rightarrow dy = \frac{dp}{kT} \cdot c$

$p^2 = \left(\frac{kT}{c}\right)^2 y^2$   $\frac{kT}{c} dy = dp$

$n = \frac{g 4\pi}{(2\pi\hbar)^3} \int_0^\infty \frac{\left(\frac{kT}{c}\right)^2 y^2}{e^y \pm 1} \left(\frac{kT}{c} dy\right) = \frac{g 4\pi}{(2\pi\hbar)^3} \left(\frac{kT}{c}\right)^3 \int_0^\infty \frac{y^2}{e^y \pm 1} dy$

$n \propto (kT)^3$

Non-Rel:  $kT \ll mc^2$  :  $E = mc^2 + \frac{p^2}{2m}$

$n = \frac{g 4\pi}{(2\pi\hbar)^3} \int_0^\infty \frac{p^2 dp}{\exp\left(\frac{(mc^2 + p^2/2m)}{kT}\right) \pm 1}$

$\frac{mc^2}{kT} \gg 1 \Rightarrow e^{\frac{mc^2}{kT}} \gg 1$

$\Rightarrow$  can ignore  $\pm 1$

$n \approx \frac{g 4\pi}{(2\pi\hbar)^3} \int_0^\infty \frac{p^2}{e^{\frac{mc^2}{kT}} e^{\frac{p^2}{2mkT}}} dp$

$\approx \frac{g 4\pi}{(2\pi\hbar)^3} e^{-\frac{mc^2}{kT}} \int_0^\infty p^2 e^{-\frac{p^2}{2mkT}} dp$

$$\bullet \quad y = \frac{p}{\sqrt{2m\kappa T}} \Rightarrow$$

$$dp = \sqrt{2m\kappa T} dy$$

$$p^2 = (2m\kappa T) y^2$$

$$\approx \frac{4\pi g}{(2\pi\hbar)^3} e^{-mc^2/\kappa T} \int_0^\infty (2m\kappa T) y^2 e^{-y^2} (\sqrt{2m\kappa T} dy)$$

$$\propto (2m\kappa T)^{3/2} e^{-mc^2/\kappa T}$$

(PS7 + 8) + B3 on exam

D.3: (a), (b)

PS7(4): (a), (b)

o: times today

$$\bullet \text{ Recall: } 1+z = \frac{R_0}{R}, \quad \frac{\lambda_0}{R_0} = \frac{\lambda}{R}, \quad H^2 = \frac{8\pi G}{3} \rho + \frac{\kappa c^2}{R^2}$$

$$\Omega_i = \rho_i / \rho_c$$

$$\rho_c: \kappa=0, \quad \frac{H^2}{8\pi G} = \rho_c$$

$$\rightarrow \rho_m = \rho_{m,0} \left(\frac{R_0}{R}\right)^3 = \rho_{m,0} (1+z)^3$$

$$\rho_r = \rho_{r,0} \left(\frac{R_0}{R}\right)^4 = \rho_{r,0} (1+z)^4$$

$$\rho_v: \text{Constant over time} \quad \rho_v = \rho_{v,0} \left(\frac{R_0}{R}\right)^0 = \rho_{v,0} (1+z)^0$$

(3) Q4:  $\rightarrow$  Find in Exam B.3 (b)

$$R_0 r_{ph}(z) = \frac{z_c}{R_0} (1+z)^{-1/2}$$

plug in  $z_{cmb} = z$

Vacuum:  $\rho_v$ : const.

$\rightarrow$  Can get into from thermo:

A.8: Sard  $\underline{w=-1}$ :  $\boxed{p = w \rho c^2}$

pressure less:

$$1^{st} \text{ law of thermo: } Tds = dE + p dV$$

$\bullet \rightarrow dS = 0$ : Entropy for Grand can is conserved

$$\bullet dE + p dV = 0$$

$\bullet$  Assume 2: Sphere:  $V = \frac{4\pi}{3} R^3$

$M_{tot}$

matter:  $w=0$

rad:  $w = 1/3$

• Assume 2:  $E = M_{tot} c^2$   $\rho = \frac{M}{Vol}$   
 $= \rho Vol c^2$   
 $= \rho \frac{4\pi}{3} R^3 c^2$

$$\frac{dE}{dt} + \rho \frac{dV}{dt} = 0$$

$$\begin{aligned} \frac{dE}{dt} &= \dot{\rho} \frac{4\pi}{3} R^3 c^2 + \rho \frac{4\pi}{3} (3R^2) \dot{R} c^2 \\ &= c^2 \frac{4\pi}{3} R^2 \left( \dot{\rho} \frac{R}{3} + \rho \dot{R} \right) \end{aligned}$$

$$\frac{dV}{dt} = 4\pi R^2 \dot{R}$$

$$4\pi c^2 R^2 \left( \dot{\rho} \frac{R}{3} + \rho \dot{R} \right) + \rho 4\pi R^2 \dot{R} = 0$$

$$c^2 \left( \dot{\rho} \frac{R}{3} + \rho \dot{R} \right) + \rho \dot{R} = 0$$

Solve for  $\dot{\rho}$

$$\frac{3}{R} \left( \dot{\rho} \frac{R}{3} + \rho \dot{R} + \frac{\rho \dot{R}}{c^2} \right) = 0$$

$$\dot{\rho} + \frac{3}{R} \dot{R} \left( \rho + \frac{\rho}{c^2} \right) = 0 \quad \leftarrow$$

→ Also: Plug into  $H^2 = \frac{8\pi G}{3} \rho - \frac{kc^2}{R^2}$

to get an acceleration

$$\ddot{R} = - \frac{4\pi G R}{3} \left( \rho + \frac{3\rho}{c^2} \right)$$

A.6:  $\rho_v = \rho_{v,0} (R_0/R)^0$  constant

Models:

• EdS:  $\Omega_{m,0} = 1, K = 0$

• Radiation:  $\rho_{r,0} \dots$

In reality all 3:

$$\rightarrow \rho_{tot}(R) = \rho_m(R) + \rho_r(R) + \rho_v$$

$$H^2 = \frac{8\pi G}{3} \rho_{tot} - \frac{kc^2}{R^2}$$

$$\tilde{\Omega} = \rho_{tot}/\rho_c, \quad \Omega_m = \frac{\rho_m}{\rho_c}$$

divide by  $\rho_c$

$$\frac{\rho_{tot}}{\rho_c} = \frac{\rho_m}{\rho_c} + \frac{\rho_r}{\rho_c} + \frac{\rho_v}{\rho_c} \quad \neq$$

$$\ast \quad \tilde{\Omega} = \Omega_m + \Omega_r + \Omega_v$$

$$\Omega_m = \frac{\rho_m}{\rho_c} = \frac{\rho_{m,0} (1+z)^3}{\rho_c} \quad \rho_c = \frac{H^2 3}{8\pi G} \quad \rho_{c,0} = \frac{H_0^2 3}{8\pi G}$$

$$\begin{cases} \Omega_m = \left( \frac{\rho_{m,0}}{\rho_{c,0}} \right) (1+z)^3 = \Omega_{m,0} (1+z)^3 \\ \Omega_r = \Omega_{r,0} (1+z)^4 \\ \Omega_v = \Omega_v \end{cases}$$

$$H^2 = \frac{8\pi G}{3} \rho_{tot} - \frac{kc^2}{R^2}$$

$$\rho_{tot} = \tilde{\Omega} \rho_c \Rightarrow H^2 = \frac{8\pi G}{3} (\tilde{\Omega} \rho_c) - \frac{kc^2}{R^2}$$

$$\rho_c = \frac{H^2 3}{8\pi G}$$

$$H^2 = \tilde{\Omega} H^2 - \frac{kc^2}{R^2}$$

$$H^2 (1 - \tilde{\Omega}) = - \frac{kc^2}{R^2}$$

$$H^2 = - \frac{kc^2}{R^2 (1 - \tilde{\Omega})} = \frac{1}{R^2} \frac{kc^2}{(\tilde{\Omega} - 1)}$$

$$R^2 = \frac{1}{H^2} \frac{kc^2}{(\tilde{\Omega} - 1)}$$

$$\begin{aligned} \text{get } R_0 \quad R(t)^2 \\ t=t_0 \quad R(t_0) = R_0 \\ R_0^2 = \frac{1}{H_0^2} \frac{kc^2}{(\tilde{\Omega} - 1)} \end{aligned}$$

↓ Solve for  $(kc^2)$

$$\ast R_0^2 H_0^2 (\tilde{\Omega} - 1) = kc^2 \ast$$

$$\hookrightarrow \text{Go back F.E.: } H^2 = \frac{8\pi G}{3} \rho - \frac{kc^2}{R^2}$$

Does not depend on when we look

$$t = t_0: \quad R_0^2 H_0^2 (\tilde{\Omega} - 1) = kc^2$$

$$H^2 = \frac{8\pi G}{3} \rho = \frac{R_0^2 H_0^2 (\tilde{\Omega} - 1)}{R^2}$$

$$H^2 = H_0^2 \left( \Omega_{m,0} \left( \frac{R_0}{R} \right)^3 + \Omega_{r,0} \left( \frac{R_0}{R} \right)^4 + \Omega_v \right) + (1 - \tilde{\Omega}) \left( \frac{R_0}{R} \right)^2$$

$$K: \sim \left( \frac{R_0}{R} \right)^2$$

$$\int \rho_0 dr = \int \frac{C dz}{H(z)} \leftarrow H(z)$$

Exam (b):

(b) optimal PS8: EdS: (matter)  $\Omega_{m,0} = 1$

$$\Omega_{v,0} = 0$$

$$\Omega_{r,0} = 0$$

$$\tilde{\Omega} = \Omega_{tot} = \Omega_{m,0}$$

$$H^2 = H_0^2 (\underbrace{\Omega_{m,0}}_1 (1+z)^3 + \underbrace{\Omega_{v,0}}_0 + \underbrace{\Omega_{r,0}}_0 + \underbrace{(1-\tilde{\Omega})}_0)$$

in EdS  $\downarrow$

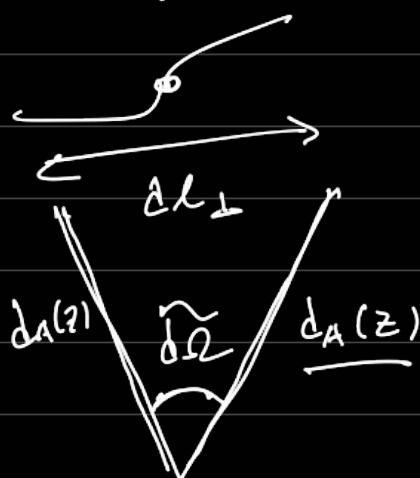
$$H^2 = H_0^2 \Omega_{m,0} (1+z)^3 \leftarrow \text{Becomes exactly Exam B.3 (b)}$$

$$= H_0^2 (1+z)^3$$

Obs. Cosmology:

Measuring angles:

$$ds^2 = \underbrace{(cdt)^2}_{\text{time}} - R^2 \left( \underbrace{dr^2}_{\text{radial}} + \underbrace{S_k^2 d\Omega^2}_{\text{angular}} \right)$$



$R$ : expansion

$$\vec{x}(t) = R(t) \vec{r}$$

An angle in cosm:  $(R S_k d\Omega) = d\tilde{\Omega}$

In Euclidean "normal" its just  $d\Omega = d\tilde{\Omega}$

$\rightarrow d_A(z) d\tilde{\Omega} = dL_{\perp}$

$s \approx r \theta$

$\uparrow \quad \uparrow$

$dA(z) \quad d\tilde{\Omega}$

$$dL_{\perp} \equiv R S_k d\Omega$$

$$dA(z) = R S_k$$

$$\frac{R_0}{R} = 1+z$$

Solve:  $R = \frac{R_0}{1+z}$

$z \gg 1$

$$d_A(z) = \frac{R_0 S_k}{(1+z)} * (\text{angular diameter distance}) \quad \left(\frac{1}{2}\right) \left(1 + \frac{1}{\sqrt{1+z}}\right)$$

$$k=0: S_k(r) = r$$

$$d_A(z) = \frac{R_0 r(z)}{(1+z)}$$

• Optimal PS7: k=0 use  $\Omega$   
 $dm = \frac{cdt}{R(t)}$

$$S_k(r) = \begin{cases} +\sinh(r) & k=1 \\ r & k=0 \\ +\sin(r) & k=-1 \end{cases}$$

B.3 (C)  $\Theta_{\text{cmb}} d_A(z) = \frac{R_0 r_A(z)}{(1+z)}$

$$\Theta_{\text{cmb}} = \frac{R_0 r_A(z)}{(1+z) d_A(z)}$$

$$R_0 r_A(z) = \frac{zc}{H_0} (1+z)^{-1/2}$$

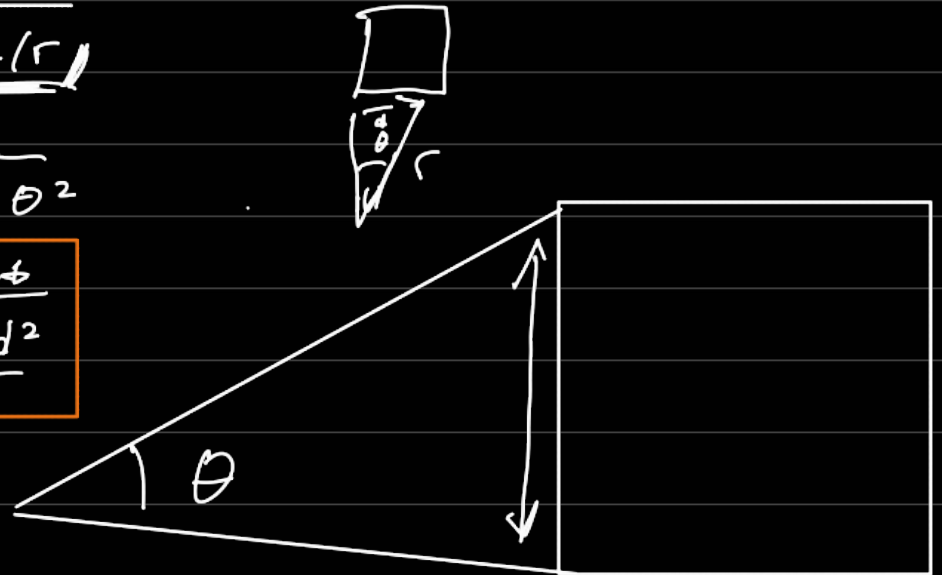
$$\Theta_{\text{cmb}} = \frac{zc}{H_0} \frac{(1+z)^{-1/2} H_0 (1+z)}{(1+z) zc} = (1+z)^{-1/2}$$

$$\Theta_{\text{cmb}} = (1+z)^{-1/2}$$

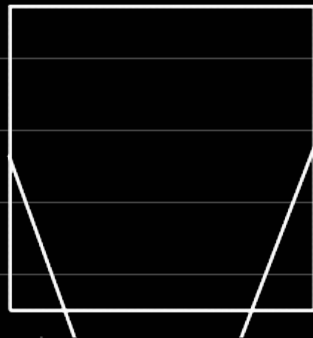
Q2 PS8:  $r\theta = d/r$

$$M_0 = \frac{F}{\theta^2} = \frac{L_{\text{tot}}}{4\pi r^2 \theta^2}$$

$$M_0 = \frac{L_{\text{tot}}}{4\pi r^2 \left(\frac{d}{r}\right)^2} = \frac{L_{\text{tot}}}{4\pi d^2}$$



$$\theta \cdot \theta = \theta^2$$





•  $d_A(z)$ : angular distance

$d_L(z)$ : luminosity distance:

$$d_A(z) = \frac{R_0 S_K}{(1+z)}$$

$$d_L(z) = R_0 S_K (1+z) *$$

(b)  $r\theta = d$  in Euclidean "normal"

but in cosm.

$$r \rightarrow d_A(z)$$

$$d_A = d/\theta \Rightarrow \hat{\theta} = d/d_A$$

• same time

Flux:

$$\tilde{F} = \frac{L_{tot}}{4\pi d_L^2}$$

$$\mu(z) = \frac{\tilde{F}}{\tilde{\theta}^2} = \left( \frac{L_{tot}}{4\pi d_L^2} \right) \left( \frac{d_A}{d} \right)^2 = \frac{L_{tot}}{4\pi d^2} \left( \frac{d_A}{d_L} \right)^2$$

$$\frac{d_L}{d_A} = (1+z)^2 \text{ or } \left( \frac{d_A}{d_L} \right)^2 = \left( \frac{1}{(1+z)^2} \right)^2 = \frac{1}{(1+z)^4}$$

$$\mu(z) = \frac{L_{tot}}{4\pi d^2} \left( \frac{1}{(1+z)^4} \right) = \mu_0 \frac{1}{(1+z)^4}$$

PS8 Q3 (a): just ask -

\* QFT:  $\langle \Omega | \Omega \rangle$ :



